Network Design and Implementation Using BGP

eBGP Routing and ACL Security for a University Network

Report

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Submission date	15 October 2026
Tool used	Cisco Packet Tracer

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1. Introduction

Modern organizations with more than one site depend on routing protocols that are scalable, controllable and secure. The Border Gateway Protocol (BGP) is the routing protocol of the Internet and is used to exchange routing information between independent networks called Autonomous Systems (AS). When BGP runs between two different autonomous systems it is called external BGP (eBGP).

This report documents the design, implementation and testing of a three-router network built in Cisco Packet Tracer. The network demonstrates BGP and eBGP for inter-site routing and an Extended Access Control List (ACL) to enforce a security policy on a protected server network.

1.1 Objectives

- Design a realistic multi-site network with its own topology and IP addressing scheme.
- Implement BGP and eBGP between three autonomous systems.
- Advertise the LAN networks over BGP so that all sites can reach each other.
- Implement an Extended ACL that restricts access to the server according to a defined security policy.
- Test and verify neighbor relationships, route advertisements, routing tables, connectivity and ACL behaviour.

2. Network Scenario

The chosen scenario is a university with a Main Campus and a Branch Campus, connected through an Internet Service Provider (ISP).

- Main Campus (AS 65001): contains a Staff LAN and a Server LAN hosting the university web server.
- ISP (AS 65002): a transit provider that carries traffic between the two campuses.
- Branch Campus (AS 65003): a remote campus with its own LAN. Branch users need to reach the university web portal but must not have unrestricted access to the server network.

2.1 Network Requirements

1. Each organization (Main Campus, ISP, Branch) operates as a separate autonomous system and exchanges routes using eBGP.
2. All LANs must be reachable from each other (PC0, PC1 and the Server).
3. Branch Campus users may only access the web server over HTTP (TCP 80) and HTTPS (TCP 443).
4. All other traffic from the Branch LAN to the Server LAN (for example ICMP ping) must be denied.
5. Main Campus Staff LAN users must have full access to the Server LAN.
6. Remote device management of the Main Campus router must be restricted to the Staff LAN.

2.2 Device Inventory

Device	Model	Purpose
Router3	Cisco 2911	Main Campus edge router (AS 65001)
Router5	Cisco 2911	ISP transit router (AS 65002)
Router4	Cisco 2911	Branch Campus edge router (AS 65003)
Switch2	Cisco 2960-24TT	Staff LAN access switch
Switch3	Cisco 2960-24TT	Server LAN access switch
Switch4	Cisco 2960-24TT	Branch LAN access switch
PC0	PC-PT	Main Campus staff workstation
Server0	Server-PT	University web server (HTTP/HTTPS)
PC1	PC-PT	Branch Campus workstation

3. Network Topology

The topology below was built in Packet Tracer. Router3 connects to the ISP router (Router5) through Gig0/0 and to two LANs through Gig0/1 and Gig0/2. Router5 connects to Router4 at the Branch Campus.

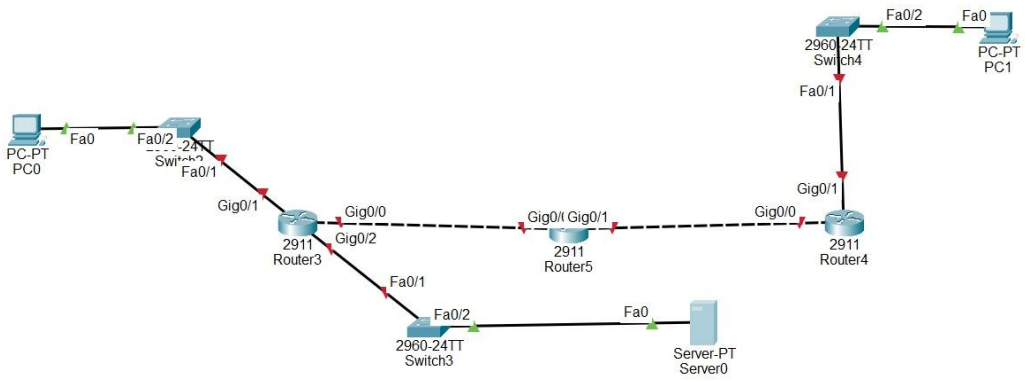


Figure 1: Network topology in Cisco Packet Tracer

3.1 Interface Connections

Device	Interface	Connects to	Remote interface
Router3	Gig0/0	Router5	Gig0/0
Router3	Gig0/1	Switch2	Fa0/1
Router3	Gig0/2	Switch3	Fa0/1

Device	Interface	Connects to	Remote interface
Router5	Gig0/1	Router4	Gig0/0
Router4	Gig0/1	Switch4	Fa0/1
Switch2	Fa0/2	PC0	Fa0
Switch3	Fa0/2	Server0	Fa0
Switch4	Fa0/2	PC1	Fa0

4. IP Addressing Scheme

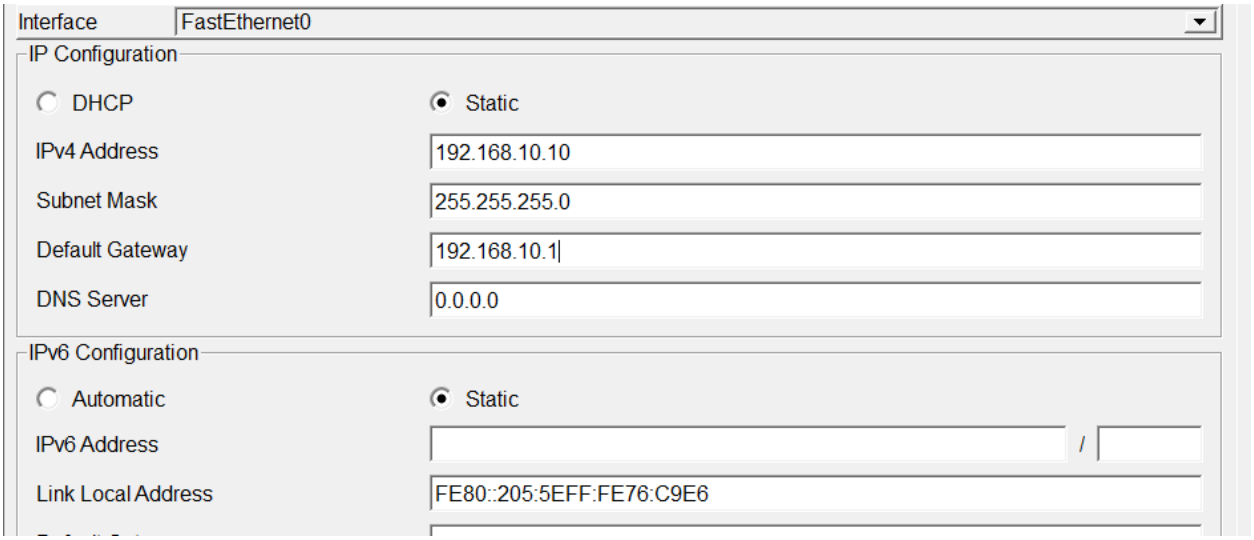
Private address space is used throughout. The LANs use /24 subnets and the point-to-point router links use /30 subnets to avoid wasting addresses.

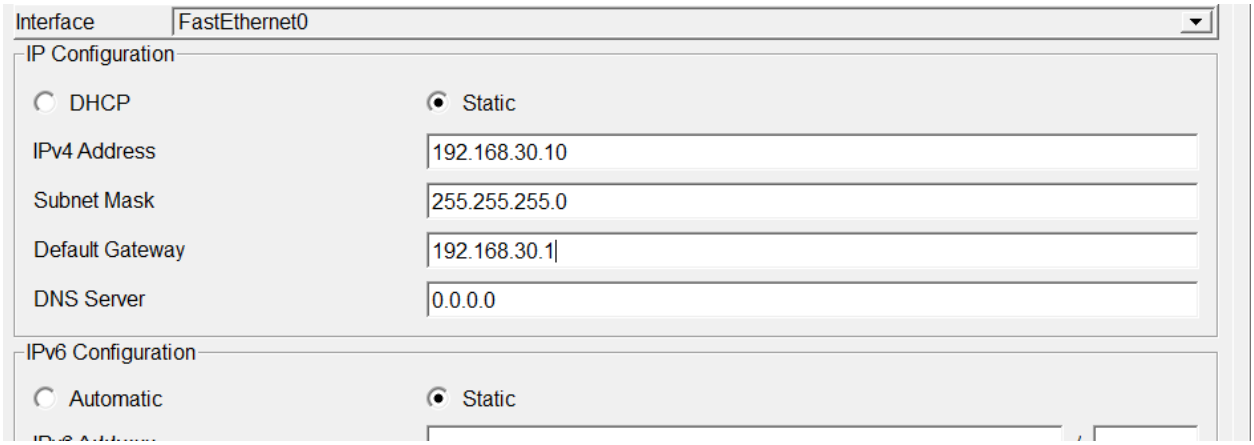
4.1 Network Addressing

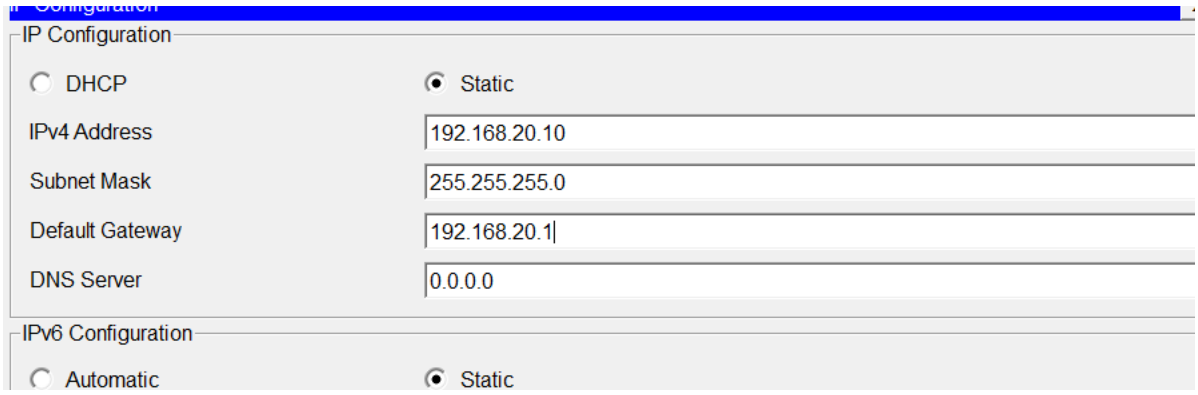
Network	Subnet	Description
Staff LAN	192.168.10.0/24	Main Campus staff (PC0)
Server LAN	192.168.20.0/24	Main Campus servers (Server0)
Branch LAN	192.168.30.0/24	Branch Campus users (PC1)
Router3 - Router5 link	10.0.35.0/30	Main Campus to ISP
Router5 - Router4 link	10.0.54.0/30	ISP to Branch Campus

4.2 Device Addressing

Device	Interface	IP address	Subnet mask	Gateway
Router3	Gig0/0	10.0.35.1	255.255.255.252	-
Router3	Gig0/1	192.168.10.1	255.255.255.0	-
Router3	Gig0/2	192.168.20.1	255.255.255.0	-
Router5	Gig0/0	10.0.35.2	255.255.255.252	-
Router5	Gig0/1	10.0.54.1	255.255.255.252	-
Router4	Gig0/0	10.0.54.2	255.255.255.252	-
Router4	Gig0/1	192.168.30.1	255.255.255.0	-
PC0	Fa0	192.168.10.10	255.255.255.0	192.168.10.1
Server0	Fa0	192.168.20.10	255.255.255.0	192.168.20.1
PC1	Fa0	192.168.30.10	255.255.255.0	192.168.30.1


Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::205:5EFF:FE76:C9E6


Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.30.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.30.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static


IP Configuration
DHCP
Static
IPv4 Address
192.168.20.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

5. BGP and eBGP Design

5.1 Autonomous System Plan

Router	AS number	Router ID	eBGP neighbor(s)
Router3	65001	1.1.1.1	Router5 (10.0.35.2, AS 65002)
Router5	65002	2.2.2.2	Router3 (10.0.35.1, AS 65001); Router4 (10.0.54.2, AS 65003)
Router4	65003	3.3.3.3	Router5 (10.0.54.1, AS 65002)

Private AS numbers (64512 - 65534) are used because this is a lab environment. Each router is in a different AS, so every BGP session in this design is an eBGP session. eBGP neighbors are formed over the directly connected /30 link addresses, so no multihop or loopback peering is needed.

5.2 Advertised Networks

Router	Networks advertised	Purpose
Router3	192.168.10.0/24, 192.168.20.0/24, 10.0.35.0/30	Staff LAN, Server LAN and its ISP link
Router5	10.0.35.0/30, 10.0.54.0/30	ISP link networks
Router4	192.168.30.0/24, 10.0.54.0/30	Branch LAN and its ISP link

Router5 acts as a transit AS. eBGP automatically re-advertises the best paths it learns from one peer to its other peers, so Router3 learns the Branch LAN and Router4 learns the Main Campus LANs through the ISP.

5.3 BGP Security Note

On physical Cisco routers each eBGP session can additionally be protected with an MD5 password using the neighbor password command. This feature is not supported by the BGP implementation in Cisco Packet Tracer, so it was not used in this lab. Security in this design is provided by the ACL on the server network and by restricting remote management access.

6. ACL Design and Implementation

6.1 Security Requirement

The Server LAN holds the university web server and must be protected. Branch Campus users only need the web portal. The ACL therefore allows Branch users to reach Server0 on HTTP and HTTPS only, and blocks all other Branch traffic to the Server LAN. Staff LAN users keep full access.

6.2 ACL Rules

Seq	Rule	Effect
1	permit tcp 192.168.30.0/24 to host 192.168.20.10 eq 80	Branch can use HTTP
2	permit tcp 192.168.30.0/24 to host 192.168.20.10 eq 443	Branch can use HTTPS
3	deny ip 192.168.30.0/24 to 192.168.20.0/24	All other Branch traffic is blocked
4	permit ip 192.168.10.0/24 to 192.168.20.0/24	Staff LAN has full access
5	permit ip any any	Allows any other traffic

6.3 ACL Placement

The ACL is named SERVER_PROTECT and is an Extended ACL. It is applied outbound on Router3 interface GigabitEthernet0/2 (towards the Server LAN). Extended ACLs are normally placed close to the source, but here the policy protects a single destination network, so placing it on the exit interface towards the servers keeps the rule set in one place and filters all sources. Because the ACL is outbound, return traffic from the server to the clients is not filtered, so permitted web sessions work normally.

6.4 Additional Access Control

Standard ACL 10 is applied to the VTY lines on Router3 so that only the Staff LAN (192.168.10.0/24) can manage the router remotely, and a VTY password is required.

7. Device Configuration

The full configurations used on each device are shown below.

7.1 Router3 (Main Campus, AS 65001)

```
enable
configure terminal
hostname Router3
no ip domain-lookup
banner motd #Authorized Access Only - Main Campus#

interface GigabitEthernet0/0
 description Link to ISP Router5
```

```
 ip address 10.0.35.1 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 description Staff LAN
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/2
 description Server LAN
 ip address 192.168.20.1 255.255.255.0
 no shutdown

router bgp 65001
 bgp router-id 1.1.1.1
 bgp log-neighbor-changes
 neighbor 10.0.35.2 remote-as 65002
 network 192.168.10.0 mask 255.255.255.0
 network 192.168.20.0 mask 255.255.255.0
 network 10.0.35.0 mask 255.255.255.252

ip access-list extended SERVER_PROTECT
 remark Branch may only use HTTP/HTTPS to Server0
 permit tcp 192.168.30.0 0.0.0.255 host 192.168.20.10 eq 80
 permit tcp 192.168.30.0 0.0.0.255 host 192.168.20.10 eq 443
 remark Block everything else from Branch
 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 remark Staff LAN has full access
 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 remark Allow any other traffic
 permit ip any any

interface GigabitEthernet0/2
 ip access-group SERVER_PROTECT out

enable secret cisco123
access-list 10 permit 192.168.10.0 0.0.0.255
line vty 0 4
 access-class 10 in
 password cisco
 login
line console 0
 password cisco
 login
 logging synchronous
service password-encryption
end
write memory
```

IOS Command Line Interface

```
Router3#enable
Router3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router3(config)#hostname Router3
Router3(config)#no ip domain-lookup
Router3(config)#no ip access-list extended SERVER_PROTECT
Router3(config)#
Router3(config)#interface GigabitEthernet0/0
Router3(config-if)# description Link to ISP Router5
Router3(config-if)# ip address 10.0.35.1 255.255.255.252
Router3(config-if)# no shutdown
Router3(config-if)#interface GigabitEthernet0/1
Router3(config-if)# description Staff LAN
Router3(config-if)# ip address 192.168.10.1 255.255.255.0
Router3(config-if)# no shutdown
Router3(config-if)#interface GigabitEthernet0/2
Router3(config-if)# description Server LAN
Router3(config-if)# ip address 192.168.20.1 255.255.255.0
Router3(config-if)# no shutdown
Router3(config-if)#exit
Router3(config)#
Router3(config)#router bgp 65001
Router3(config-router)# bgp router-id 1.1.1.1
Router3(config-router)# bgp log-neighbor-changes
Router3(config-router)# neighbor 10.0.35.2 remote-as 65002
Router3(config-router)# network 192.168.10.0 mask 255.255.255.0
Router3(config-router)# network 192.168.20.0 mask 255.255.255.0
Router3(config-router)# network 10.0.35.0 mask 255.255.255.252
Router3(config-router)#exit
Router3(config)#
Router3(config)#ip access-list extended SERVER_PROTECT
Router3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 192.168.20.10 eq 80
Router3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 192.168.20.10 eq 443
Router3(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
Router3(config-ext-nacl)# permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router3(config-ext-nacl)# permit ip any any
Router3(config-ext-nacl)#exit
Router3(config)#
Router3(config)#interface GigabitEthernet0/2
Router3(config-if)# ip access-group SERVER_PROTECT out
```

```
Router3(config-ext-nacl)#exit
Router3(config)#
Router3(config)#interface GigabitEthernet0/2
Router3(config-if)# ip access-group SERVER_PROTECT out
Router3(config-if)#exit
Router3(config)#
Router3(config)#enable secret cisco123
Router3(config)#access-list 10 permit 192.168.10.0 0.0.0.255
Router3(config)#line vty 0 4
Router3(config-line)# access-class 10 in
Router3(config-line)# password cisco
Router3(config-line)# login
Router3(config-line)#exit
Router3(config)#line console 0
Router3(config-line)# password cisco
Router3(config-line)# login
Router3(config-line)# logging synchronous
Router3(config-line)#exit
Router3(config)#service password-encryption
Router3(config)#end
```

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7.2 Router5 (ISP, AS 65002)

```
enable
configure terminal
hostname Router5
no ip domain-lookup

interface GigabitEthernet0/0
 description Link to Main Campus Router3
 ip address 10.0.35.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 description Link to Branch Router4
 ip address 10.0.54.1 255.255.255.252
 no shutdown

router bgp 65002
 bgp router-id 2.2.2.2
 bgp log-neighbor-changes
 neighbor 10.0.35.1 remote-as 65001
 neighbor 10.0.54.2 remote-as 65003
 network 10.0.35.0 mask 255.255.255.252
 network 10.0.54.0 mask 255.255.255.252

enable secret cisco123
line console 0
 password cisco
 login
 logging synchronous
service password-encryption
end
write memory
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router5
Router5(config)#no ip domain-lookup
Router5(config)#
Router5(config)#interface GigabitEthernet0/0
Router5(config-if)# description Link to Main Campus Router3
Router5(config-if)# ip address 10.0.35.2 255.255.255.252
Router5(config-if)# no shutdown

Router5(config-if)#interface GigabitEthernet0/1
Router5(config-if)# description Link to Branch Router4
Router5(config-if)# ip address 10.0.54.1 255.255.255.252
Router5(config-if)# no shutdown

Router5(config-if)#
Router5(config-if)#router bgp 65002
Router5(config-router)# bgp router-id 2.2.2.2
Router5(config-router)# bgp log-neighbor-changes
Router5(config-router)# neighbor 10.0.35.1 remote-as 65001
Router5(config-router)# neighbor 10.0.54.2 remote-as 65003
Router5(config-router)# network 10.0.35.0 mask 255.255.255.252
Router5(config-router)# network 10.0.54.0 mask 255.255.255.252
Router5(config-router)#
Router5(config-router)#enable secret cisco123
Router5(config)#line console 0
Router5(config-line)# password cisco
Router5(config-line)# login
Router5(config-line)# logging synchronous
Router5(config-line)#service password-encryption
Router5(config)#end
Router5#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

7.3 Router4 (Branch Campus, AS 65003)

```
enable
configure terminal
hostname Router4
no ip domain-lookup

interface GigabitEthernet0/0
 description Link to ISP Router5
 ip address 10.0.54.2 255.255.255.252
 no shutdown
interface GigabitEthernet0/1
 description Branch LAN
 ip address 192.168.30.1 255.255.255.0
 no shutdown

router bgp 65003
 bgp router-id 3.3.3.3
 bgp log-neighbor-changes
 neighbor 10.0.54.1 remote-as 65002
 network 192.168.30.0 mask 255.255.255.0
```

```
 network 10.0.54.0 mask 255.255.255.252

enable secret cisco123
line console 0
 password cisco
 login
 logging synchronous
service password-encryption
end
write memory
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router4
Router4(config)#no ip domain-lookup
Router4(config)#
Router4(config)#interface GigabitEthernet0/0
Router4(config-if)# description Link to ISP Router5
Router4(config-if)# ip address 10.0.54.2 255.255.255.252
Router4(config-if)# no shutdown

Router4(config-if)#interface GigabitEthernet0/1
Router4(config-if)# description Branch LAN
Router4(config-if)# ip address 192.168.30.1 255.255.255.0
Router4(config-if)# no shutdown

Router4(config-if)#
Router4(config-if)#router bgp 65003
Router4(config-router)# bgp router-id 3.3.3.3
Router4(config-router)# bgp log-neighbor-changes
Router4(config-router)# neighbor 10.0.54.1 remote-as 65002
Router4(config-router)# network 192.168.30.0 mask 255.255.255.0
Router4(config-router)# network 10.0.54.0 mask 255.255.255.252
Router4(config-router)#
Router4(config-router)#enable secret cisco123
Router4(config)#line console 0
Router4(config-line)# password cisco
Router4(config-line)# login
Router4(config-line)# logging synchronous
Router4(config-line)#service password-encryption
Router4(config)#end
Router4#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

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7.4 Switches

```
enable
configure terminal
hostname Switch4
no ip domain-lookup
enable secret cisco123
line console 0
 password cisco
 login
end
write memory
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch2
Switch2(config)#no ip domain-lookup
Switch2(config)#enable secret cisco123
Switch2(config)#line console 0
Switch2(config-line)# password cisco
Switch2(config-line)# login
Switch2(config-line)#end
Switch2#
%SYS-5-CONFIG_I: Configured from console by console
```

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```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch3
Switch3(config)#no ip domain-lookup
Switch3(config)#enable secret cisco123
Switch3(config)#line console 0
Switch3(config-line)# password cisco
Switch3(config-line)# login
Switch3(config-line)#end
Switch3#
%SYS-5-CONFIG_I: Configured from console by console
```

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```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Switch4
Switch4(config)#no ip domain-lookup
Switch4(config)#enable secret cisco123
Switch4(config)#line console 0
Switch4(config-line)# password cisco
Switch4(config-line)# login
Switch4(config-line)#end
Switch4#
%SYS-5-CONFIG_I: Configured from console by console
```

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7.5 End Device Configuration

- PC0: IP 192.168.10.10, mask 255.255.255.0, gateway 192.168.10.1
- Server0: IP 192.168.20.10, mask 255.255.255.0, gateway 192.168.20.1; HTTP and HTTPS services turned on
- PC1: IP 192.168.30.10, mask 255.255.255.0, gateway 192.168.30.1

8. Testing and Verification

This section describes each test, the command used and the expected result. Paste the Packet Tracer screenshot into each placeholder.

8.1 Verify eBGP Neighbor Relationships

Run the following on Router3, Router5 and Router4. The neighbor state must show a prefix count (Established), not Idle or Active.

```
show ip bgp summary
show ip bgp neighbors
```

IOS Command Line Interface

```
Router3#show ip bgp summary
BGP router identifier 1.1.1.1, local AS number 65001
BGP table version is 7, main routing table version 6
6 network entries using 792 bytes of memory
6 path entries using 312 bytes of memory
3/2 BGP path/bestpath attribute entries using 460 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1668 total bytes of memory
BGP activity 5/0 prefixes, 6/0 paths, scan interval 60 secs

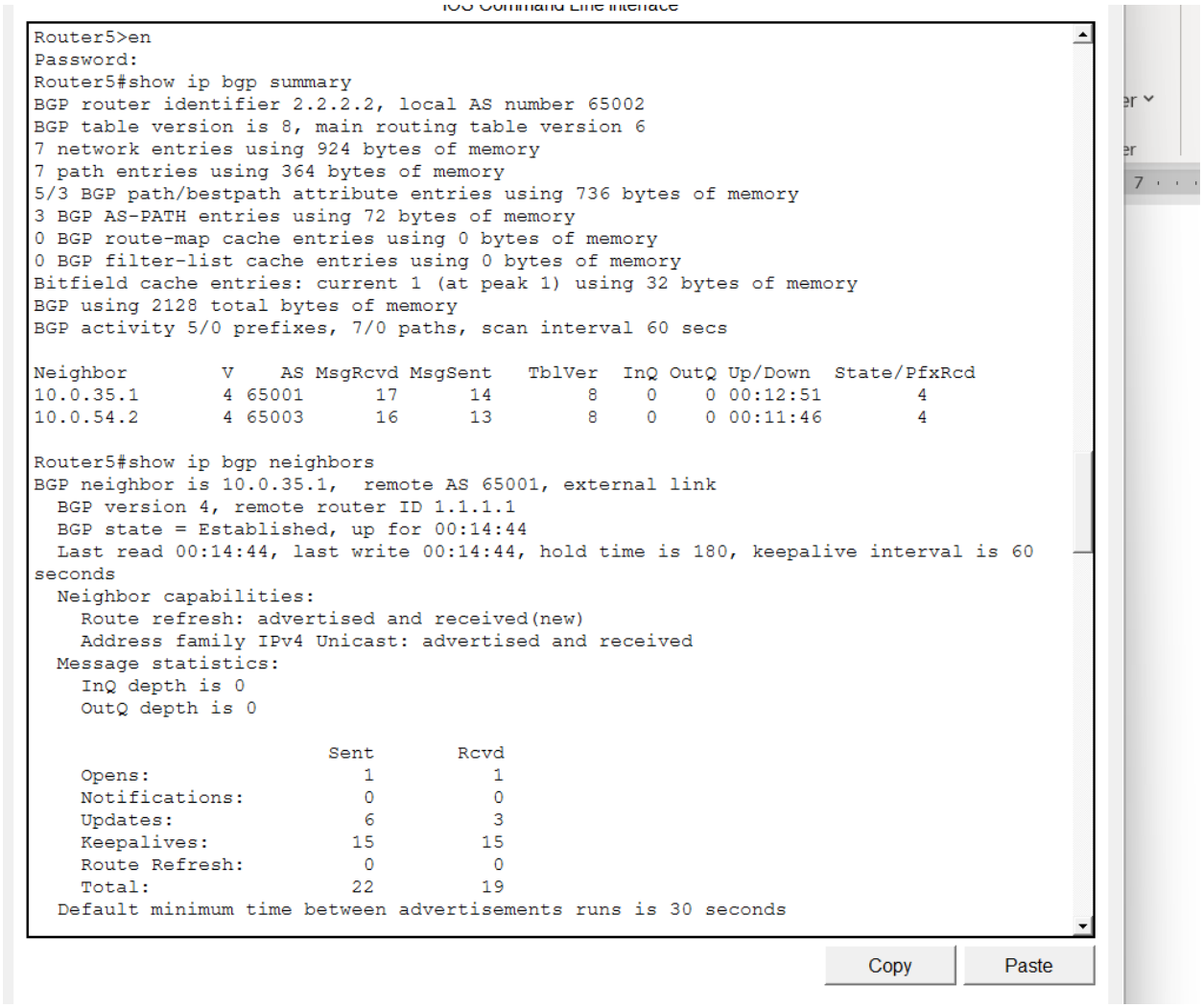
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.35.2       4 65002      19      13        7    0    0 00:11:14        4

Router3#show ip bgp neighbors
BGP neighbor is 10.0.35.2,  remote AS 65002, external link
  BGP version 4, remote router ID 2.2.2.2
  BGP state = Established, up for 00:13:43
  Last read 00:13:43, last write 00:13:43, hold time is 180, keepalive interval is 60
seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                3          6
    Keepalives:            14         14
    Route Refresh:          0          3
    Total:                 18         24
  Default minimum time between advertisements runs is 30 seconds

 For address family: IPv4 Unicast
  BGP table version 7, neighbor version 6/0
  Output queue size : 0
```

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IOS Command Line Interface
Router5>en
Password:
Router5#show ip bgp summary
BGP router identifier 2.2.2.2, local AS number 65002
BGP table version is 8, main routing table version 6
7 network entries using 924 bytes of memory
7 path entries using 364 bytes of memory
5/3 BGP path/bestpath attribute entries using 736 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 2128 total bytes of memory
BGP activity 5/0 prefixes, 7/0 paths, scan interval 60 secs
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd
10.0.35.1 4 65001 17 14 8 0 0 00:12:51 4
10.0.54.2 4 65003 16 13 8 0 0 00:11:46 4
Router5#show ip bgp neighbors
BGP neighbor is 10.0.35.1, remote AS 65001, external link
BGP version 4, remote router ID 1.1.1.1
BGP state = Established, up for 00:14:44
Last read 00:14:44, last write 00:14:44, hold time is 180, keepalive interval is 60
seconds
Neighbor capabilities:
Route refresh: advertised and received(new)
Address family IPv4 Unicast: advertised and received
Message statistics:
InQ depth is 0
OutQ depth is 0
Sent Rcvd
Opens: 1 1
Notifications: 0 0
Updates: 6 3
Keepalives: 15 15
Route Refresh: 0 0
Total: 22 19
Default minimum time between advertisements runs is 30 seconds
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IOS Command Line Interface

```
Router4#show ip bgp summary
BGP router identifier 3.3.3.3, local AS number 65003
BGP table version is 7, main routing table version 6
6 network entries using 792 bytes of memory
6 path entries using 312 bytes of memory
4/3 BGP path/bestpath attribute entries using 644 bytes of memory
3 BGP AS-PATH entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1852 total bytes of memory
BGP activity 5/0 prefixes, 6/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.54.1       4 65002      21      16        7    0    0 00:14:59        4

Router4#show ip bgp neighbors
BGP neighbor is 10.0.54.1,  remote AS 65002, external link
  BGP version 4, remote router ID 2.2.2.2
  BGP state = Established, up for 00:15:20
  Last read 00:15:20, last write 00:15:20, hold time is 180, keepalive interval is 60
seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                3          5
    Keepalives:            16         16
    Route Refresh:          0          1
    Total:                 20         23
  Default minimum time between advertisements runs is 30 seconds

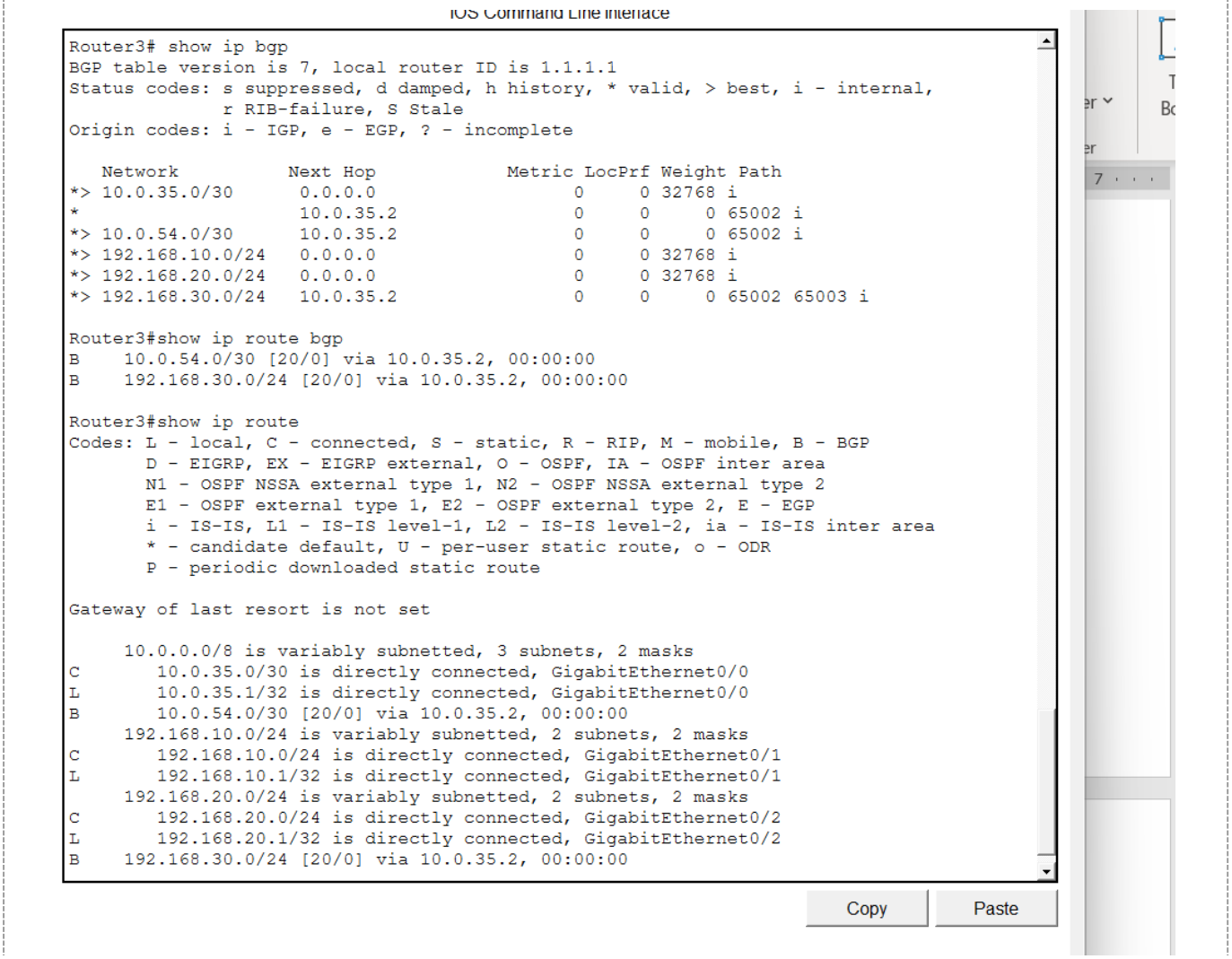
 For address family: IPv4 Unicast
  BGP table version 7, neighbor version 6/0
```

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8.2 Verify Route Advertisements and Routing Tables

```
show ip bgp
show ip route bgp
show ip route
```

- Router3 should have a BGP (B) route to 192.168.30.0/24.
- Router4 should have BGP (B) routes to 192.168.10.0/24 and 192.168.20.0/24.
- Router5 should have BGP routes to all three LANs.


IOS Command Line Interface
Router3# show ip bgp
BGP table version is 7, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
*> 10.0.35.0/30 0.0.0.0 0 0 32768 i
* 10.0.35.2 0 0 0 65002 i
*> 10.0.54.0/30 10.0.35.2 0 0 0 65002 i
*> 192.168.10.0/24 0.0.0.0 0 0 32768 i
*> 192.168.20.0/24 0.0.0.0 0 0 32768 i
*> 192.168.30.0/24 10.0.35.2 0 0 0 65002 65003 i
Router3#show ip route bgp
B 10.0.54.0/30 [20/0] via 10.0.35.2, 00:00:00
B 192.168.30.0/24 [20/0] via 10.0.35.2, 00:00:00
Router3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C 10.0.35.0/30 is directly connected, GigabitEthernet0/0
L 10.0.35.1/32 is directly connected, GigabitEthernet0/0
B 10.0.54.0/30 [20/0] via 10.0.35.2, 00:00:00
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/1
L 192.168.10.1/32 is directly connected, GigabitEthernet0/1
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, GigabitEthernet0/2
L 192.168.20.1/32 is directly connected, GigabitEthernet0/2
B 192.168.30.0/24 [20/0] via 10.0.35.2, 00:00:00
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```
Router5#show ip bgp
BGP table version is 8, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.35.0/30     0.0.0.0                  0     0 32768 i
*                   10.0.35.1                0     0     0 65001 i
*> 10.0.54.0/30     0.0.0.0                  0     0 32768 i
*                   10.0.54.2                0     0     0 65003 i
*> 192.168.10.0/24  10.0.35.1                0     0     0 65001 i
*> 192.168.20.0/24  10.0.35.1                0     0     0 65001 i
*> 192.168.30.0/24  10.0.54.2                0     0     0 65003 i

Router5#show ip route bgp
B    192.168.10.0/24 [20/0] via 10.0.35.1, 00:00:00
B    192.168.20.0/24 [20/0] via 10.0.35.1, 00:00:00
B    192.168.30.0/24 [20/0] via 10.0.54.2, 00:00:00

Router5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.35.0/30 is directly connected, GigabitEthernet0/0
L       10.0.35.2/32 is directly connected, GigabitEthernet0/0
C       10.0.54.0/30 is directly connected, GigabitEthernet0/1
L       10.0.54.1/32 is directly connected, GigabitEthernet0/1
B    192.168.10.0/24 [20/0] via 10.0.35.1, 00:00:00
B    192.168.20.0/24 [20/0] via 10.0.35.1, 00:00:00
B    192.168.30.0/24 [20/0] via 10.0.54.2, 00:00:00

Router5#
```

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IOS Command Line Interface

```
Router4#show ip bgp
BGP table version is 7, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 10.0.35.0/30      10.0.54.1                0     0      0 65002 i
*> 10.0.54.0/30      0.0.0.0                  0     0 32768 i
*                    10.0.54.1                0     0      0 65002 i
*> 192.168.10.0/24   10.0.54.1                0     0      0 65002 65001 i
*> 192.168.20.0/24   10.0.54.1                0     0      0 65002 65001 i
*> 192.168.30.0/24   0.0.0.0                  0     0 32768 i

Router4#show ip route bgp
B    10.0.35.0/30 [20/0] via 10.0.54.1, 00:00:00
B    192.168.10.0/24 [20/0] via 10.0.54.1, 00:00:00
B    192.168.20.0/24 [20/0] via 10.0.54.1, 00:00:00

Router4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
B       10.0.35.0/30 [20/0] via 10.0.54.1, 00:00:00
C       10.0.54.0/30 is directly connected, GigabitEthernet0/0
L       10.0.54.2/32 is directly connected, GigabitEthernet0/0
B    192.168.10.0/24 [20/0] via 10.0.54.1, 00:00:00
B    192.168.20.0/24 [20/0] via 10.0.54.1, 00:00:00
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/1
L       192.168.30.1/32 is directly connected, GigabitEthernet0/1

Router4#
```

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8.3 Connectivity Tests

Source	Test	Expected result
PC0	ping 192.168.30.10 (PC1)	Success
PC1	ping 192.168.10.10 (PC0)	Success
PC0	ping 192.168.20.10 (Server0)	Success
PC0	Web browser: http://192.168.20.10	Web page loads
PC1	Web browser: http://192.168.20.10	Web page loads (HTTP permitted)
PC1	ping 192.168.20.10 (Server0)	Fails - ACL blocks ICMP

Note: the first ping in Packet Tracer can time out while ARP completes. Repeat the ping to confirm

PC0:

```
C:\>
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125
Reply from 192.168.30.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Pc1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time=1ms TTL=125
Reply from 192.168.10.10: bytes=32 time=1ms TTL=125
Reply from 192.168.10.10: bytes=32 time=1ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Pc0 to server:

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=2ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Pc0:

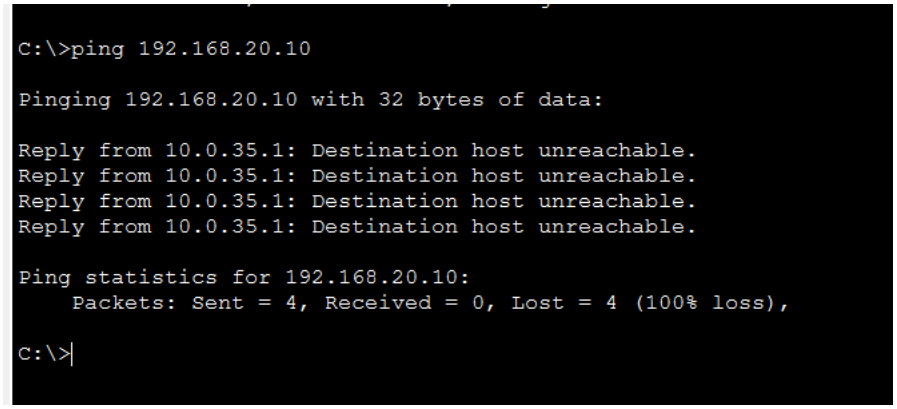


8.4 Verify the ACL

```
show access-lists
show ip interface GigabitEthernet0/2
```

After running the tests in 8.3, the match counters on Router3 should increase: the deny rule counts the blocked pings from PC1, and the permit tcp rules count the HTTP and HTTPS requests from PC1. The show ip interface output confirms that SERVER_PROTECT is applied outbound on Gig0/2.

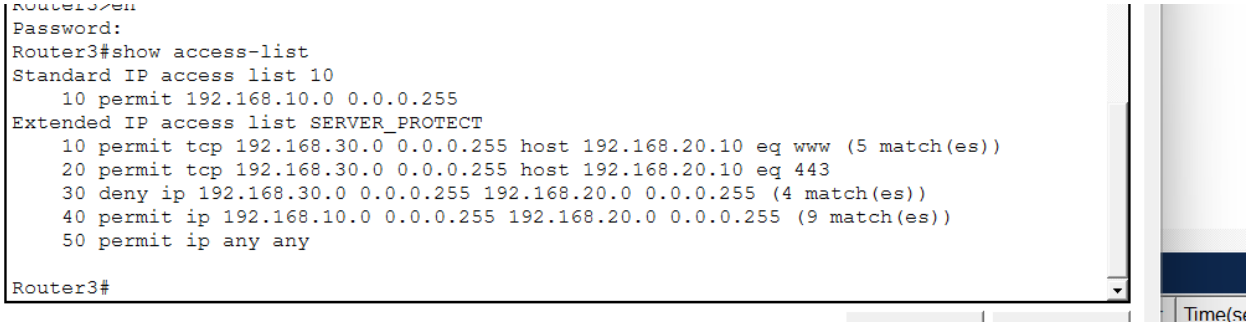
[Insert screenshot: PC1 ping to Server0 failing (Request timed out)]



[Insert screenshot: PC1 web browser successfully opening Server0]



[Insert screenshot: show access-lists on Router3 with match counters]



9. Results Summary

Requirement	Status	Evidence
eBGP neighbors established (3 routers)	Met	show ip bgp summary
LAN networks advertised and learned via BGP	Met	show ip route bgp
End-to-end connectivity between all LANs	Met	Ping tests
Branch limited to HTTP/HTTPS on server	Met	Web test and failed ping
Staff LAN has full server access	Met	Ping and web tests from PC0
Restricted remote management (VTY ACL and password)	Met	Running configuration

Update the Status column according to your actual Packet Tracer results.

10. Conclusion

The network successfully demonstrates BGP and eBGP between three autonomous systems representing a university Main Campus, an ISP and a Branch Campus. Routes for all LANs are exchanged dynamically, giving full connectivity between sites, and the Extended ACL enforces the security policy by letting the Branch reach only the web service on the server while the Staff LAN retains full access.

Possible future improvements include adding a second ISP link for redundancy with BGP path attributes (local preference, MED), prefix lists and route maps to filter advertisements, and an iBGP design with additional routers inside an AS.

11. References

- Cisco Systems. Configuring BGP, Cisco IOS IP Routing: BGP Configuration Guide.
- Cisco Systems. Configuring IP Access Lists, Cisco IOS Security Configuration Guide.